

Overview: Concrete-Pictorial-Abstract (CPA)

The Concrete-Pictorial-Abstract approach moves learning through three stages: first handling real objects (concrete), then representing them visually (pictorial), and finally using symbols like numbers and letters (abstract). This progression ensures pupils build genuine understanding rather than memorising procedures they don't comprehend. Particularly powerful for pupils who struggle with abstract concepts, it provides the foundation that makes symbols meaningful.

When to Use This Strategy

- Can follow procedures but can't explain why they work
- Gets confused when problems look slightly different
- Struggles to apply learning to new contexts
- Relies heavily on memorised rules without understanding
- Cannot draw or represent what a problem means

Step-by-Step Guide

1 Identify the concept and gather manipulatives

Before teaching, identify the core concept and select appropriate concrete materials. For maths: counters, cubes, base-ten blocks, fraction pieces. For science: actual objects, models, specimens.

Tip: Keep manipulatives organised and accessible. Have enough for the pupil to work independently.

2 Introduce with concrete materials

Let the pupil physically handle and manipulate objects. Don't rush to show them "the way" - let them explore. Guide them to discover patterns and relationships through the materials.

Teacher says: "Let's use these cubes to understand what's happening. Show me 3 groups of 4 cubes."

Tip: Resist the urge to jump to numbers. The concrete stage builds the mental model.

3 Check concrete understanding

Before moving on, ensure the pupil truly understands at the concrete level. Can they explain what they're doing? Can they solve similar problems with the manipulatives?

Teacher says: "Explain to me what you just did with the cubes. What did you find out?"

4 Move to pictorial representation

Have the pupil draw what they did with the concrete materials. The drawing should directly represent the physical objects - circles for counters, squares for cubes.

Teacher says: "Now draw what you made with the cubes. Show me 3 groups of 4 in your drawing."

Tip: Keep the manipulatives visible so they can refer back while drawing.

5 Connect pictorial to concrete

Explicitly link the drawing to the physical materials. Point to the drawing and the objects. The pupil should see they represent the same thing.

Teacher says: "Point to the cubes. Now point to your drawing. They show the same thing, don't they?"

6 Introduce abstract symbols

Only now introduce numbers and symbols. Write them next to the pictorial representation. The pupil should see how $3 \times 4 = 12$ connects to their drawing and the cubes.

Teacher says: "We can write this as 3 times 4 equals 12. Three groups of four. See how it matches your drawing?"

Tip: Keep all three representations visible: objects, drawing, and symbols side by side.

7 Practice at abstract level with option to return

Give abstract problems but keep manipulatives and drawing materials available. If the pupil gets stuck, they can return to pictorial or concrete to rebuild understanding.

Teacher says: "Try this one: 2×5 . If you need to, you can use the cubes or draw it."

English Example: Understanding sentence structure

Year 8 pupil working at KS1 level. " Learning to build simple sentences with subject-verb-object.

Concrete: Build sentences with word cards

Teacher: Give the pupil physical word cards: nouns (dog, cat, boy), verbs (chased, ate, saw), objects (ball, food, bird). Say: "Pick up a person or animal. Then pick what they did. Then pick what they did it to."

Pupil: Physically picks up cards: "The dog" + "chased" + "the ball". Arranges them in a row on the desk.

Concrete: Feel the sentence structure

Teacher: Say: "Touch each card as you read. Who? Did what? To what?" Point to each position.

Pupil: Touches each card while reading. Physically experiences the three parts of the sentence.

Pictorial: Draw the sentence

Teacher: Say: "Draw three boxes. In box 1, draw who. In box 2, draw what they did. In box 3, draw what they did it to."

Pupil: Draws dog in box 1, running action in box 2, ball in box 3. Writes the words underneath.

Abstract: Write the sentence

Teacher: Say: "Now write it as a proper sentence. Capital letter at the start, full stop at the end." Point to drawing as guide.

Pupil: Writes "The dog chased the ball." Can see how the sentence matches the pictures and the original cards.

Maths Example: Understanding multiplication as groups

Year 5 pupil working at KS1 level. " Learning 4×3 .

Concrete: Build with cubes

Teacher: Place a pile of cubes on the desk. Say: "We're going to make 4 groups. Each group needs 3 cubes. Can you make that?"

Pupil: Physically separates cubes into 4 groups of 3. Counts to check each group.

Concrete: Find the total

Teacher: Say: "How many cubes do you have altogether? Count them all."

Pupil: Counts all cubes. Discovers there are 12.

Pictorial: Draw the groups

Teacher: Give paper and pencil. Say: "Now draw your groups. Draw exactly what you made with the cubes."

Pupil: Draws 4 circles with 3 squares in each (or similar representation).

Abstract: Write the calculation

Teacher: Say: "We can write this as a number sentence. 4 groups of 3. That's 4 times 3." Write $4 \times 3 = 12$ next to the drawing.

Pupil: Sees how the symbols connect to the drawing and the cubes. Copies the number sentence.

Science Example: Understanding particle arrangement in states of matter

Year 8 pupil working at KS2 level. " Learning solid, liquid, gas particle arrangement.

Concrete: Role-play particles

Teacher: Use small objects (marbles, balls, or even pupils standing). Say: "These are particles. In a solid, they're packed tight and can't move. Pack them together."

Pupil: Arranges objects tightly together in a fixed pattern. Cannot move them without disturbing others.

Concrete: Show liquid and gas

Teacher: Say: "In a liquid, particles can slide past each other but stay close. Show me." Then: "In a gas, they spread out and move freely."

Pupil: Rearranges objects to show sliding (liquid) and scattered (gas). Physically experiences the difference.

Pictorial: Draw particle diagrams

Teacher: Give paper with three boxes labelled Solid, Liquid, Gas. Say: "Draw circles to show how the particles are arranged in each."

Pupil: Draws packed circles (solid), close but not fixed (liquid), spread out (gas). Uses concrete experience as reference.

Abstract: Use scientific terminology

Teacher: Say: "Scientists describe these as: fixed positions and vibrating (solid), close together but moving (liquid), far apart and fast (gas)." Write keywords.

Pupil: Labels diagrams with scientific vocabulary. Connects abstract words to drawings and physical experience.